Unit II

CHAPTER 6

CONTROL STRUCTURES



Learning Objectives

After studying this lesson, students will be able to:

- To gain knowledge on the various flow of control in Python language.
- To learn through the syntax how to use conditional construct to improve the efficiency of the program flow.
- To apply iteration structures to develop code to repeat the program segment for specific number of times or till the condition is satisfied.

6.1 Introduction

Programs may contain set of statements. These statements are the executable segments that yield the result. In general, statements are executed sequentially, that is the statements are executed one after another. There may be situations in our real life programming where we need to skip a segment or set of statements and execute another segment based on the test of a condition. This is called ***alternative*** or ***branching***. Also, we may need to execute a set of statements multiple times, called ***iteration*** or ***looping***. In this chapter we are to focus on the various control structures in Python, their syntax and learn how to develop the programs using them.

6.2 Control Structures

A program statement that causes a jump of control from one part of the program to another is called **control structure** or **control statement.** As you have already learnt in C++, these control statements are compound statements used to alter the control flow of the process or program depending on the state of the process.

12th Computer Science_EM Chapter 6.indd 67 26-12-2022 16:44:45